

1. What is SQL and What are its Uses?

SQL (Structured Query Language) is a domain-specific language used to store, retrieve, manipulate, and manage data in a Relational Database Management System (RDBMS).

SQL is mainly used for handling structured data, where data is organized into tables consisting of rows and columns. SQL is generally a non-case-sensitive language.

Uses of SQL:

- **SQL can retrieve data from a database.**
- **SQL can insert new data into a database.**
- **SQL can create new databases.**
- **SQL can create new tables in databases.**
- **SQL can update existing data in databases.**
- **SQL can delete data from databases.**
- **SQL can perform queries related to databases.**
- **SQL can create and manage database objects such as tables, views, and indexes.**
- **SQL can set permissions on tables, procedures, views, and other database objects.**

SQL Standards:

SQL is an American National Standards Institute (ANSI) and International Organization for Standardization (ISO) standard.

Although SQL is an ANSI/ISO standard, different database management systems may have different versions and extensions of SQL. However, most of them support the major SQL commands, such as SELECT, UPDATE, DELETE, INSERT, and WHERE, in a similar manner.

2. Different Commands of SQL, Their Uses and Syntax

The following are some of the commonly used SQL commands:

1. SELECT Command

Use:

The SELECT command is used to retrieve data from a database table.

Syntax:

SELECT column1, column2, ...

FROM table_name;

To select all columns:

SELECT *

FROM table_name;

2. UPDATE Command

Use:

The UPDATE command is used to modify existing data in a database table.

Syntax:

UPDATE table_name

SET column1 = value1, column2 = value2, ...

WHERE condition;

3. DELETE Command

Use:

The DELETE command is used to delete existing records from a database table.

Syntax:

DELETE FROM table_name

WHERE condition;

4. INSERT INTO Command

Use:

The INSERT INTO command is used to insert new records into a database table.

Syntax:

INSERT INTO table_name (column1, column2, column3, ...)

VALUES (value1, value2, value3, ...);

5. CREATE DATABASE Command

Use:

The **CREATE DATABASE** command is used to create a new database.

Syntax:

CREATE DATABASE [IF NOT EXISTS] database_name

[CHARACTER SET charset_name]

[COLLATE collation_name];

6. ALTER DATABASE Command

Use:

The **ALTER DATABASE** command is used to modify the properties or settings of an existing database.

Syntax:

ALTER DATABASE [database_name]

[DEFAULT] CHARACTER SET charset_name

[DEFAULT] COLLATE collation_name;

7. CREATE TABLE Command

Use:

The **CREATE TABLE** command is used to create a new table in a database.

Syntax:

CREATE TABLE table_name (

column1_name data_type(size) [constraints],

```
column2_name data_type(size) [constraints],  
  
...  
  
[table_constraints]  
  
);
```

8. ALTER TABLE Command

Use:

The ALTER TABLE command is used to modify the structure of an existing table. It can be used to add, modify, or remove columns and constraints.

Syntax:

```
ALTER TABLE table_name  
  
action1,  
  
action2,  
  
...;
```

9. DROP TABLE Command

Use:

The DROP TABLE command is used to permanently delete a table and all its data from a database.

Syntax:

DROP TABLE [IF EXISTS] table_name;

10. CREATE INDEX Command

Use:

The CREATE INDEX command is used to create an index on one or more columns. An index can improve the speed of data retrieval operations.

Syntax:

**CREATE [UNIQUE | FULLTEXT | SPATIAL] INDEX index_name
ON table_name (column1 [ASC | DESC], column2 [ASC | DESC], ...);**

11. DROP INDEX Command

Use:

The DROP INDEX command is used to remove an existing index from a table.

Syntax:

DROP INDEX index_name ON table_name;